AOU
الجامعة العربية المفتوحة
Arab Open University

Faculty of Business Studies

FIN340-F

ASSET PRICING AND CORPORATE FINANCE

Version B

Final Examination

2025-2026 / Spring

15 – June – 2026

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Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed: 2 hours**

Instructions:

1- Take **FIVE (5)** minutes to read all instructions carefully before attempting to write.
2- Mobile phones must be in "SWITCHED **OFF"** mode throughout the examination.
3- This is a **CLOSED** book examination. NO documents of any kind are allowed.
4- You **MUST** answer the questions using only the answer booklets provided
5- You **MUST** return the examination paper together with your answer booklets.
6- Do **NOT** write your answers in pencil except graphs/sketches
7- Students are **NOT** allowed to use any type of dictionary or mobile phones during the examination
8- Students are allowed to use ***non-programmable*** calculators only (if needed).
9- Any form of cheating will **NOT** be tolerated and will subject you to AOU Cheating Policy.

Answer ALL 5 Questions

- **Question 1: (20 marks)**

A mother plans to send her child to college for four years starting 18 years from now. Having set aside money for tuition, she decides to plan for room and board also. She estimates these costs at $20,000 per year, payable at the beginning of each year, by the time her child goes to college. If she starts next year and makes 17 payments into a savings account paying 5 percent annually, what annual payments must she make?

- **Question 2: (30 marks)**

Johnson Tire Distributors has debt with both a face and a market value of $13,000. This debt has a coupon rate of 7 percent and pays interest annually. The expected earnings before interest and taxes are $3,000, the tax rate is 40 percent, and the unlevered cost of capital is 12 percent. What is the firm's cost of equity? What is the firm's WACC? What do you conclude from the WACC you calculated?

- **Question 3: (20 marks)**

Handy Equipment has 70,000 bonds outstanding that are selling at par. Bonds with similar characteristics are yielding 7 percent. The company also has 850,000 shares of 8 percent preferred stock and 3 million shares of common stock outstanding. The preferred stock sells for $70 a share. The common stock has a beta of 1.34 and sells for $50 a share. The U.S. Treasury bill is yielding 3 percent and the return on the market is 12 percent. The corporate tax rate is 35 percent. What is the firm's weighted average cost of capital?

- **Question 4: (15 marks)**

OSINT Inc. desires a weighted average cost of capital of 12 percent. The firm has an after-tax cost of debt of 10 percent and a cost of equity of 15 percent. What debt-equity ratio is needed for the firm to achieve its targeted weighted average cost of capital? Given the debt to equity ratio you have calculated, what does it signify?

- **Question 5: (15 marks)**

Explain the direct and indirect costs of bankruptcy.

End of Questions

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Formulas

Present value: $\mathbf{PV = FV / (1 + r)^t}$

Future value: $\mathbf{FV = PV \times (1 + r)^t}$

$$\text{Annuity present value} = C \times \left(\frac{1 - \text{Present value factor}}{r}\right)$$
$$= C \times \left(\frac{1 - [1/(1+r)^t]}{r}\right)$$

$$\text{Annuity FV factor} = (\text{Future value factor} - 1)/r$$
$$= [(1+r)^t - 1]/r$$

Value of the unlevered firm= $V_U = \frac{EBIT - Taxes}{R_U} = \frac{EBIT\ X\ (1-T_C)}{R_U}$

Value of the levered firm= $V_L = V_U + T_C \times D$

$V_E = V_L - D$

The cost of equity= $R_E = R_U + (R_U - R_D) \times (D/E) \times (1 - T_C)$

Cost of equity: $R_E = R_A + (R_A - R_D) \times (D/E)$

The cost of equity using the dividend growth model approach: $\boldsymbol{R_e = D_1 / P_0 + g}$

Cost of Preferred Stock= D/Po

Weighted average cost of capital= $WACC = (E/V) \times R_E + (D/V) \times R_D$

Weighted Average Cost of Capital (WACC): $\boldsymbol{(E/V) \times R_E + (P/V) \times R_P + (D/V) \times R_D \times (1 - T_C)}$, where R_P is the cost of preferred stock.

Weighted Average cost of capital= $WACC = (E/V) \times R_E + (D/V) \times R_D \times (1 - T_C)$

Debt-equity ratio= total debt/total equity

Security Market Line (SML) Approach: $R_E = R_f + \beta_E(E(R_M) - R_f)$

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